Kernix 0.11.0 / Performance brief

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

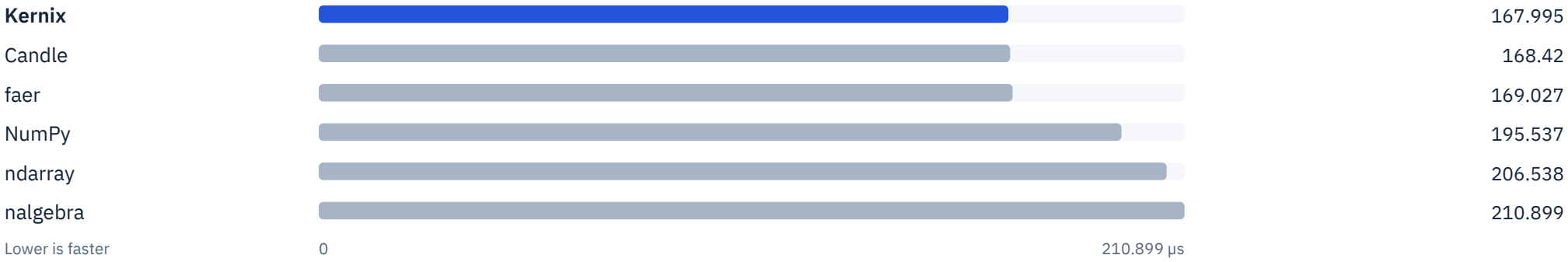
35 workloads

119 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



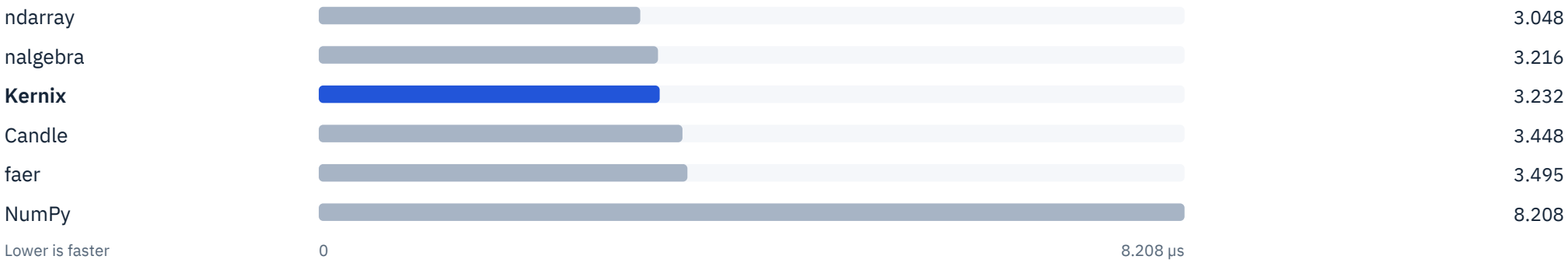
Kernix 0.11.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



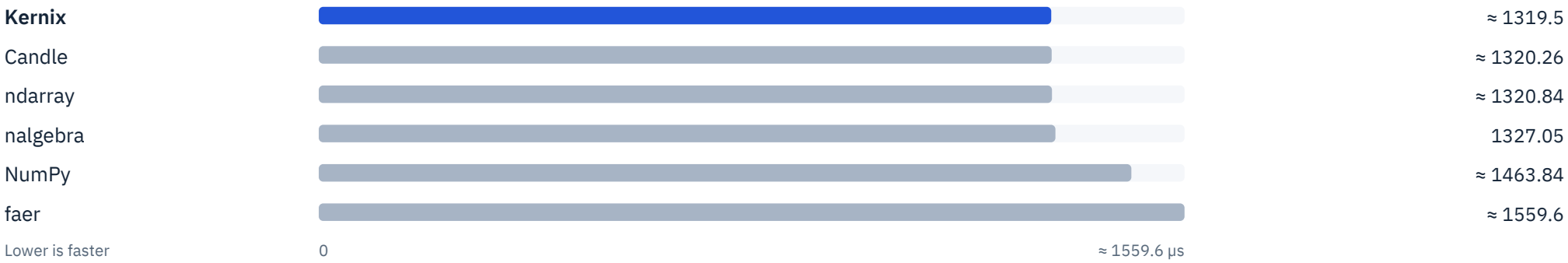
Kernix 0.11.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

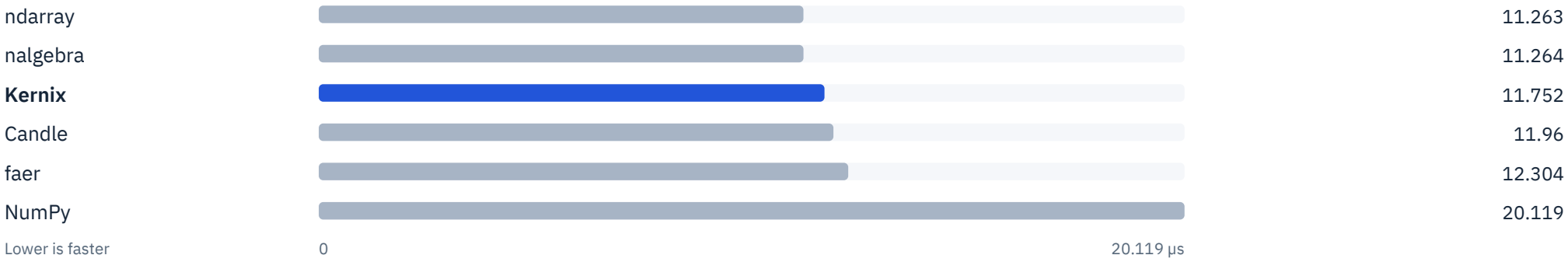
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



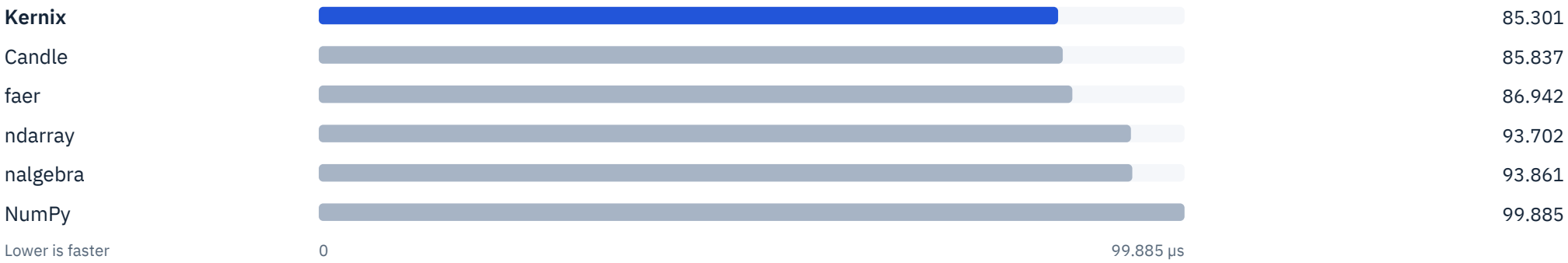
Kernix 0.11.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

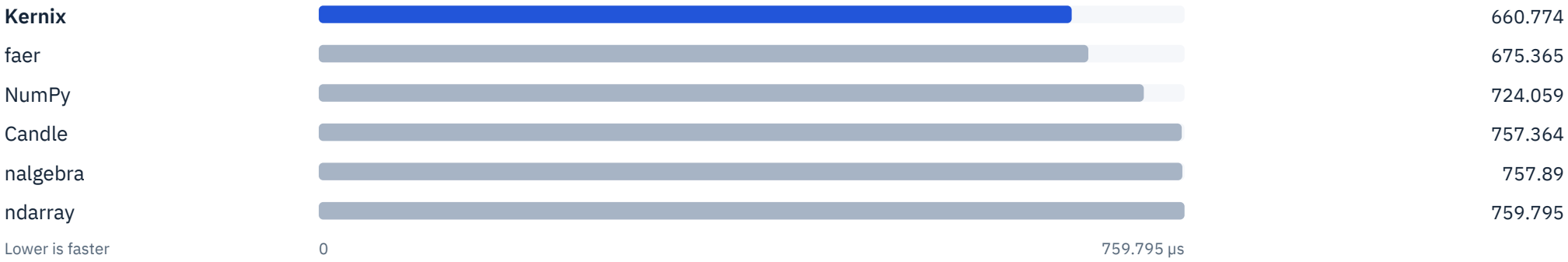
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.11.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

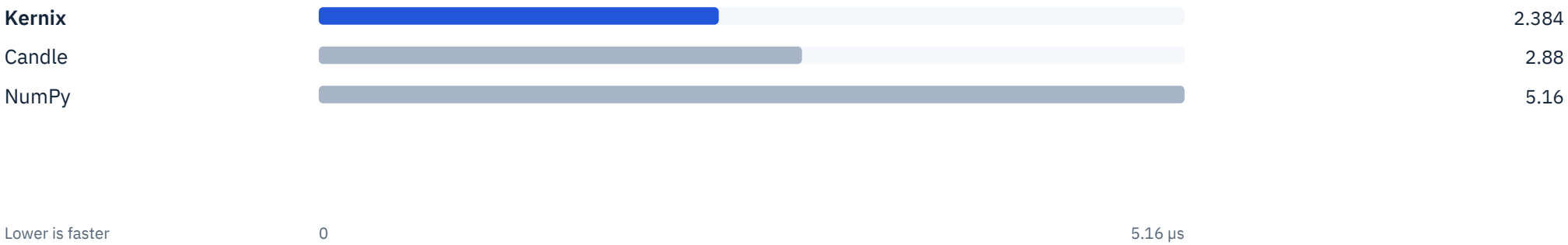
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded

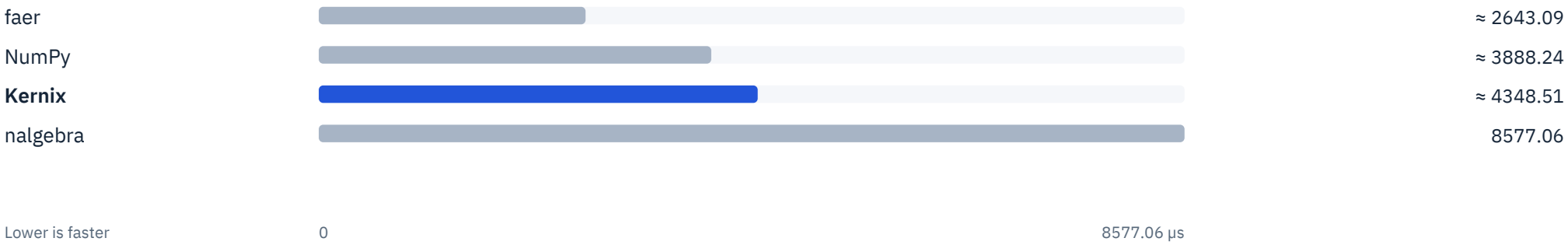


Kernix 0.11.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

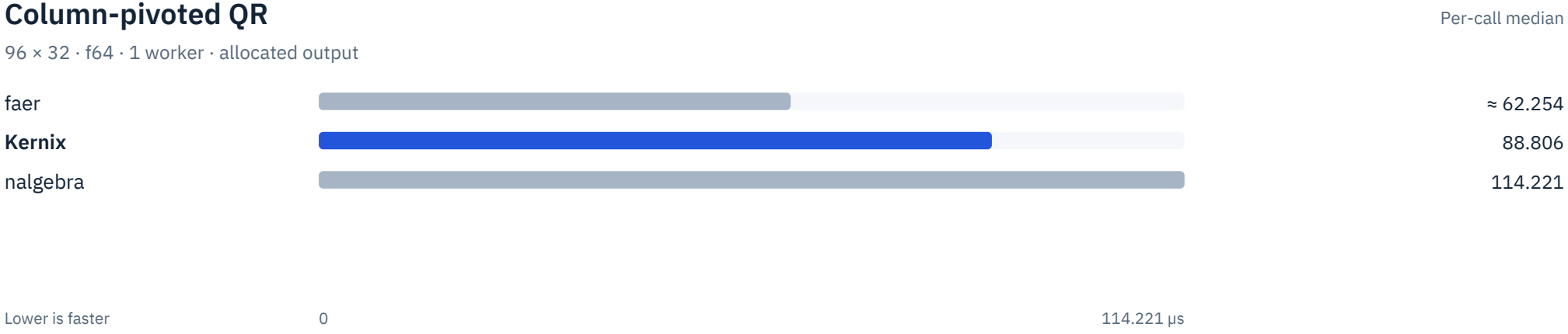
Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output



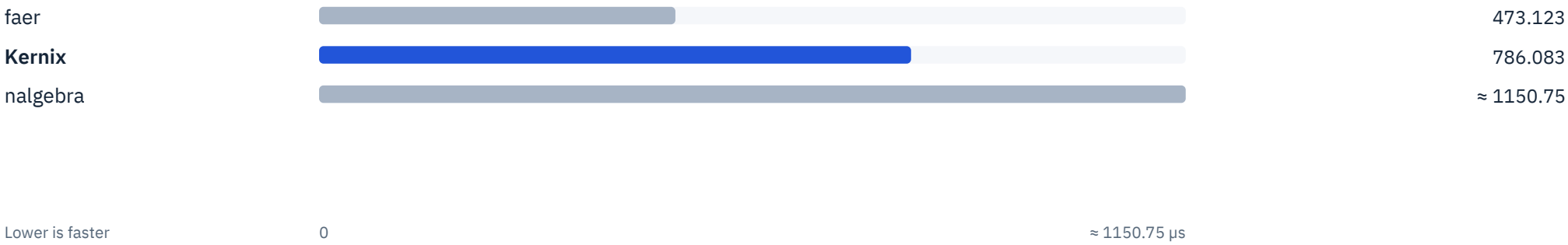
Kernix 0.11.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

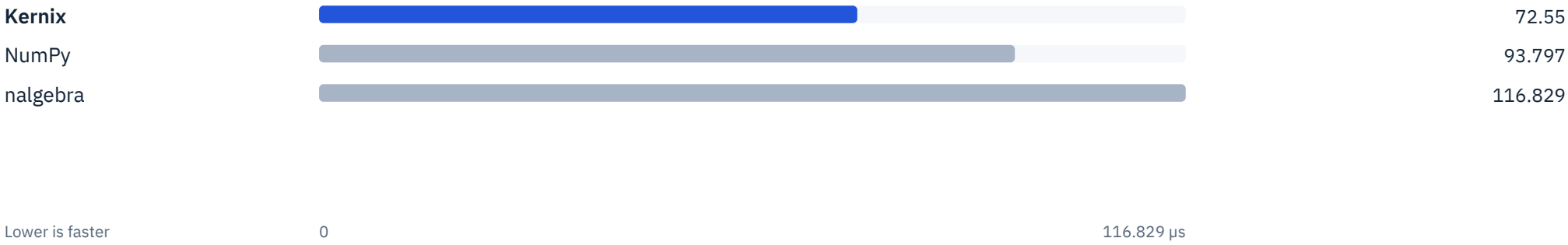
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



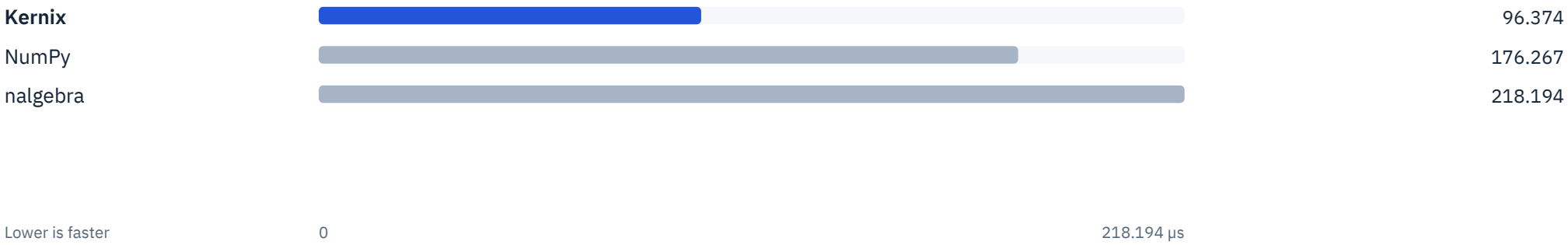
Kernix 0.11.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



Thin QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

256 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



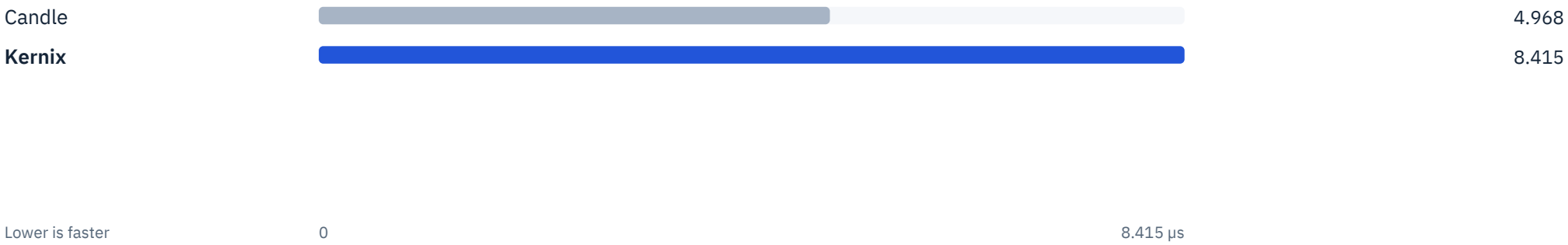
Kernix 0.11.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



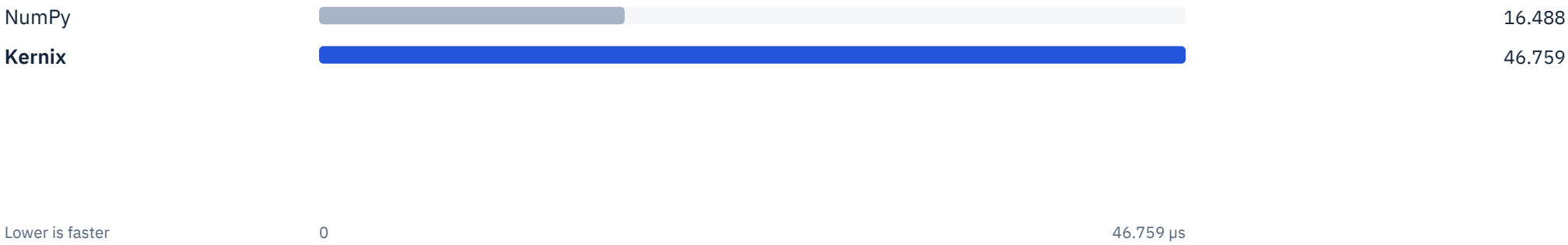
Kernix 0.11.0 / Tensor contractions

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median

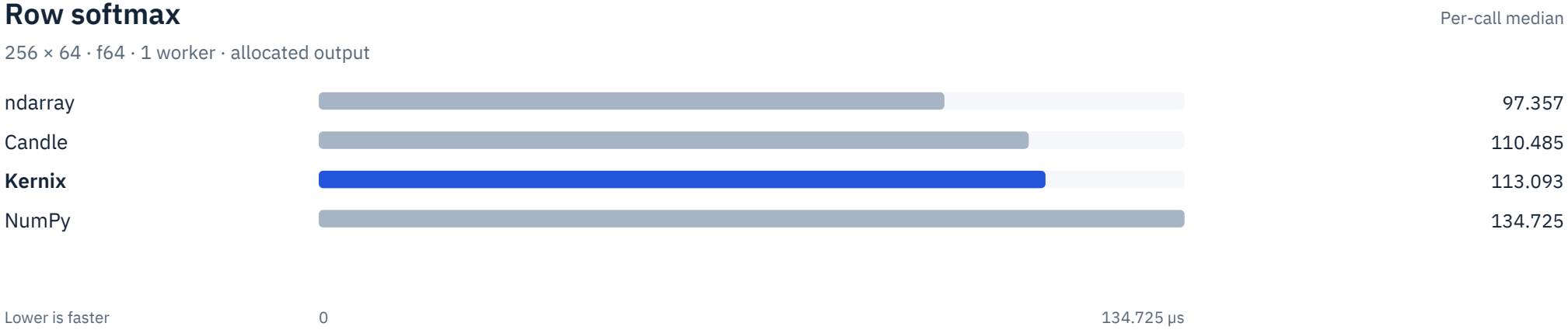


Kernix 0.11.0 / Statistics & activations

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

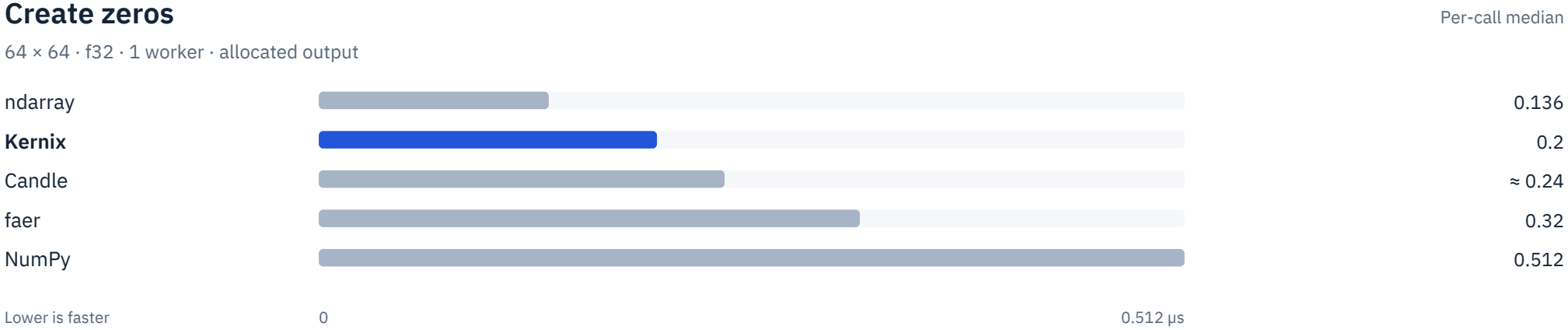


Kernix 0.11.0 / Tensor construction

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	196274	196.274
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	168795	168.795
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	99885	99.885
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	85582	85.582
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	3448	3.448
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	3048	3.048
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	3216	3.216
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	3495	3.495
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	8208	8.208
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	3232	3.232

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	21823	21.823
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	31063	31.063
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	22527	22.527
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	22648	22.648
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	22464	22.464
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	22247	22.247
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	195537	195.537
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	206538	206.538
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	167995	167.995
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	169027	169.027

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	210899	210.899
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	168420	168.42
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	1463844	1463.844
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	1320265	1320.265
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	1327050	1327.05
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1320836	1320.836
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1559604	1559.604
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	1319501	1319.501
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	11263	11.263
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	20119	20.119

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	12304	12.304
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	11752	11.752
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	11960	11.96
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	11264	11.264
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	93861	93.861
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	86942	86.942
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	85837	85.837
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	93702	93.702
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	85301	85.301
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	99885	99.885

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	757890	757.89
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	724059	724.059
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	759795	759.795
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	675365	675.365
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	660774	660.774
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	757364	757.364
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4448	4.448
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	12975	12.975
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3863.9999999999995	3.8639999999999995
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2880	2.88

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	5160	5.16
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	2384	2.384
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	16472	16.472
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	92141	92.141
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	700219	700.219
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	48895	48.895
Cholesky factorization / nalgebra 512 × 512 · f64	8577060	8577.06
Cholesky factorization / NumPy 512 × 512 · f64	3888242	3888.242
Cholesky factorization / faer 512 × 512 · f64	2643087	2643.087
Cholesky factorization / Kernix 512 × 512 · f64	4348512	4348.512

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 96 × 32 · f64	62254.00000000001	62.25400000000001
Column-pivoted QR / nalgebra 96 × 32 · f64	114221	114.221
Column-pivoted QR / Kernix 96 × 32 · f64	88806	88.806
Column-pivoted QR / nalgebra 256 × 64 · f64	1150751	1150.751
Column-pivoted QR / faer 256 × 64 · f64	473123	473.123
Column-pivoted QR / Kernix 256 × 64 · f64	786083	786.083
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	116829	116.829
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	93797	93.797
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	72550	72.55
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	176267	176.267

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	96374	96.374
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	218194	218.194
Thin QR / faer 96 × 32 · f64	65358.99999999999	65.35899999999999
Thin QR / NumPy 96 × 32 · f64	66574	66.574
Thin QR / Kernix 96 × 32 · f64	71790	71.79
Thin QR / nalgebra 96 × 32 · f64	61327.00000000001	61.32700000000001
Thin QR / faer 256 × 64 · f64	446563	446.563
Thin QR / nalgebra 256 × 64 · f64	585640	585.64
Thin QR / Kernix 256 × 64 · f64	695382	695.382
Thin QR / NumPy 256 × 64 · f64	569305	569.305

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	27287	27.287
Cholesky factorization / faer 64 × 64 · f64	13208	13.208
Cholesky factorization / NumPy 64 × 64 · f64	18088	18.088
Cholesky factorization / nalgebra 64 × 64 · f64	20431	20.431
Cholesky factorization / NumPy 128 × 128 · f64	84206	84.206
Cholesky factorization / nalgebra 128 × 128 · f64	129037	129.037
Cholesky factorization / faer 128 × 128 · f64	64942	64.942
Cholesky factorization / Kernix 128 × 128 · f64	280737	280.737
Cholesky factorization / NumPy 256 × 256 · f64	748650	748.65
Cholesky factorization / faer 256 × 256 · f64	405645	405.645

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 256 × 256 · f64	1037404	1037.404
Cholesky factorization / Kernix 256 × 256 · f64	768276	768.276
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4968	4.968
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	8415	8.415
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7200	7.2
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3504	3.504
Model inference / Candle batched normalized/sliced MLP · f64	5368	5.368
Model inference / Kernix batched normalized/sliced MLP · f64	3071	3.071
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3824	3.824
Model recording + forward / Candle batched normalized/sliced MLP · f64	7655.999999999999	7.655999999999999

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Kernix prepared graph → fresh gradients · f64	4184	4.184
Model backward / Candle prepared graph → fresh gradients · f64	21408	21.408
Parameter update / Kernix transactional SGD, fixed gradients · f64	728	0.728
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	46759	46.759
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	16488	16.488
Row softmax / Candle 256 × 64 · f64	110485	110.485
Row softmax / ndarray 256 × 64 · f64	97357	97.357
Row softmax / Kernix 256 × 64 · f64	113093	113.093
Row softmax / NumPy 256 × 64 · f64	134725	134.725
Create zeros / faer 64 × 64 · f32	320	0.32

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Candle 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create zeros / NumPy 64 × 64 · f32	512	0.512
Create zeros / Kernix 64 × 64 · f32	200	0.2
Create zeros / ndarray 64 × 64 · f32	136	0.136
Create ones / Candle 64 × 64 · f32	344	0.344
Create ones / ndarray 64 × 64 · f32	208	0.208
Create ones / NumPy 64 × 64 · f32	1528	1.528
Create ones / Kernix 64 × 64 · f32	256	0.256
Create ones / faer 64 × 64 · f32	239.99999999999997	0.23999999999999997